Module 4 — Core Organic Chemistry

4.1 Basic Concepts

Nomenclature

Organic compounds are named using the IUPAC system. The name is built from: (1) the **prefix** indicating substituents and their positions, (2) the **root name** from the longest continuous carbon chain, and (3) the **suffix** indicating the principal functional group. Number the chain to give the lowest possible locant to the principal functional group.

Carbon Atoms	Prefix	Carbon Atoms	Prefix
1	meth-	6	hex-
2	eth-	7	hept-
3	prop-	8	oct-
4	but-	9	non-
5	pent-	10	dec-

Suffixes indicate the functional group: **-ane** (alkane), **-ene** (alkene), **-ol** (alcohol), **-al** (aldehyde), **-one** (ketone), **-oic acid** (carboxylic acid), **-oate** (ester), **-amine** (amine), **-amide** (amide), **-nitrile** (nitrile).

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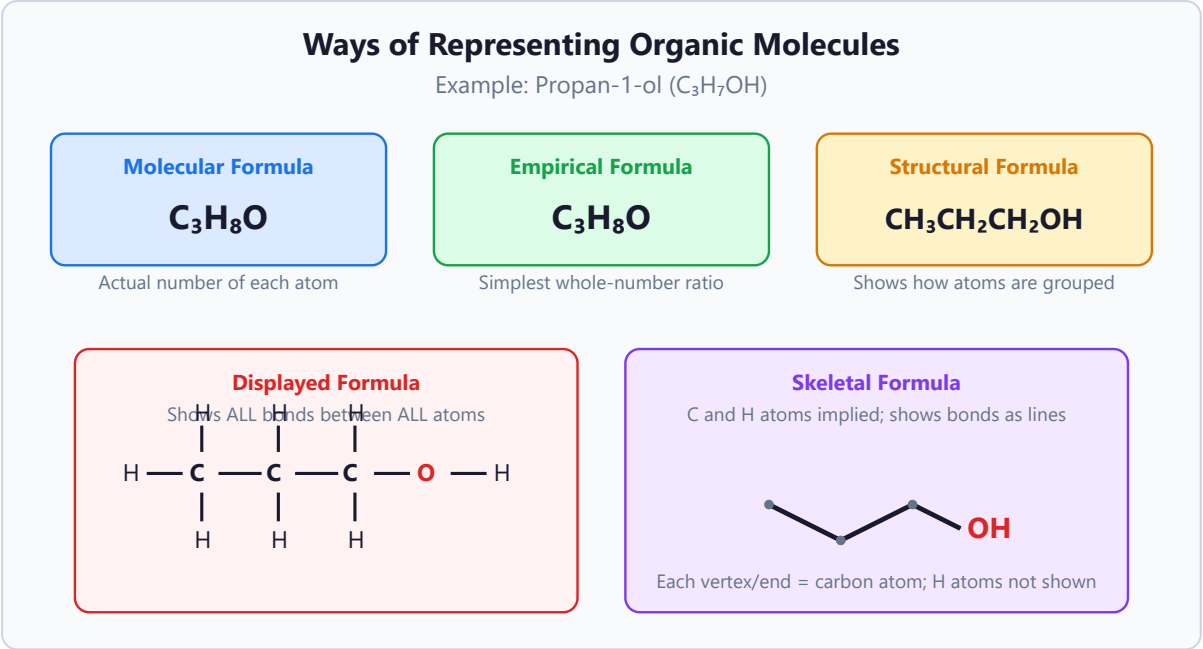

Functional Groups and Homologous Series

Functional group: An atom or group of atoms responsible for the characteristic chemical reactions of a compound. Compounds with the same functional group belong to the same **homologous series** — a family of compounds with the same functional group, the same general formula, and similar chemical properties. Successive members differ by CH_2.

Homologous Series	Functional Group	General Formula	Example
Alkane	C–C and C–H only (saturated)	C_nH_{2n+2}	Ethane, C_2H_6
Alkene	C=C (carbon-carbon double bond)	C_nH_{2n}	Propene, C_3H_6
Alcohol	–OH (hydroxyl group)	$C_nH_{2n+1}OH$	Ethanol, C_2H_5OH
Haloalkane	–X (halogen: F, Cl, Br, I)	$C_nH_{2n+1}X$	Chloroethane, C_2H_5Cl
Aldehyde	–CHO (terminal carbonyl)	$C_nH_{2n}O$	Ethanal, CH_3CHO
Ketone	>C=O (non-terminal carbonyl)	$C_nH_{2n}O$	Propanone, CH_3COCH_3

Carboxylic acid	–COOH	$C_nH_{2n}O_2$	Ethanoic acid, CH_3COOH
Ester	–COO–	Variable	Ethyl ethanoate, $CH_3COOC_2H_5$

Types of Isomerism

Structural isomers: Molecules with the same molecular formula but different structural arrangements of atoms. Three types:

• **Chain isomerism** — different carbon skeleton (e.g. butane vs methylpropane)

• **Position isomerism** — same skeleton but functional group at a different position (e.g. butan-1-ol vs butan-2-ol)

• **Functional group isomerism** — different functional groups (e.g. propanal vs propanone; ethanol vs methoxymethane)

Stereoisomers: Molecules with the same structural formula but a different arrangement of atoms in three-dimensional space.

• **E/Z isomerism** (geometric isomerism) occurs in alkenes when there is restricted rotation around the C=C bond AND each carbon of the double bond has two different substituents.

• **Optical isomerism** occurs when a carbon atom is bonded to four different groups (a chiral centre), producing non-superimposable mirror images called enantiomers.

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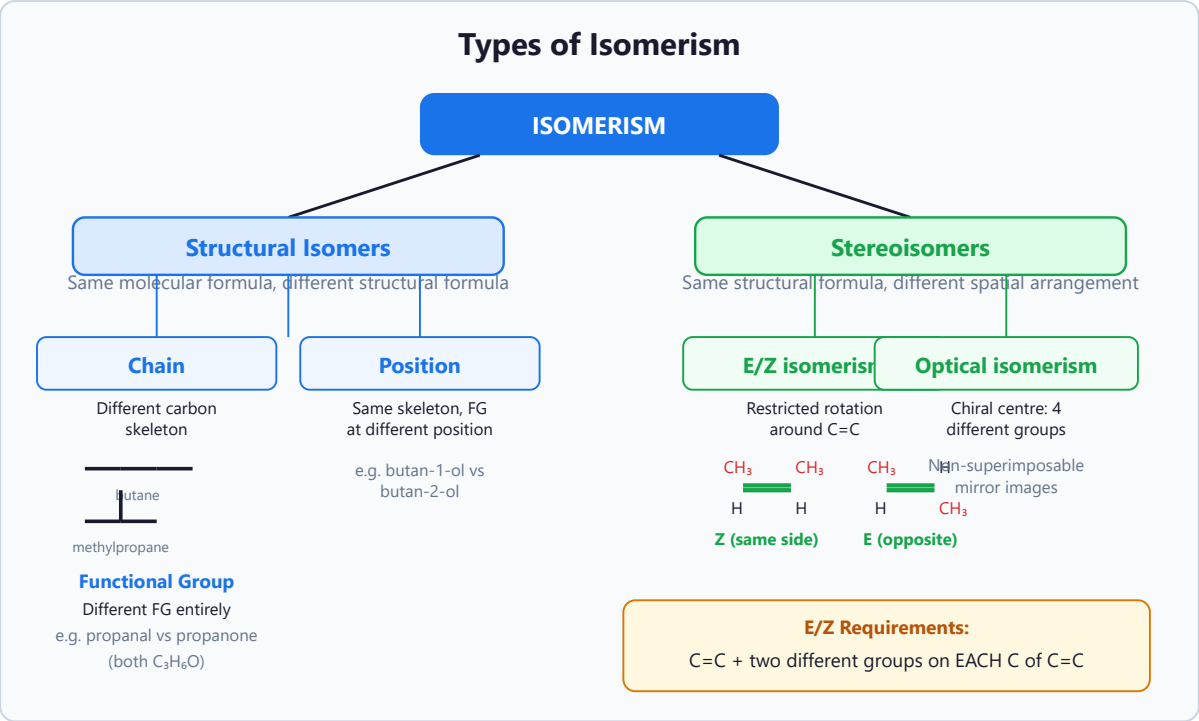

E/Z isomers are distinguished using **Cahn-Ingold-Prelog (CIP) priority rules** based on atomic number of atoms directly bonded to the C=C carbons. The atom with the higher atomic number has higher priority. Z (zusammen, "together") has the higher priority groups on the same side; E (entgegen, "opposite") has them on opposite sides.

Exam Tip: For E/Z isomerism, you need **both** conditions: (1) restricted rotation (C=C double bond), AND (2) two different groups on each carbon of the double bond. If either carbon of the C=C has two identical substituents, E/Z isomers cannot exist. For CIP rules, if the first atoms are the same, look at the next atoms along the chain.

4.2 Alkanes

Alkanes: Saturated hydrocarbons with only single C–C and C–H bonds. They are relatively unreactive due to the high bond enthalpy of C–C (347 kJ mol^{-1}) and C–H (413 kJ mol^{-1}) bonds and their non-polar nature (C and H have similar electronegativities, so the bonds have very small dipoles).

Combustion

Complete combustion (excess oxygen) produces CO_2 and H_2O. This is the basis of using hydrocarbons as fuels:

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l) \quad \Delta H = -890 \text{ kJ mol}^{-1}$$

Incomplete combustion (limited oxygen) produces CO and/or C (soot) along with H_2O. Carbon monoxide is toxic because it binds irreversibly to haemoglobin (forming carboxyhaemoglobin), reducing the blood's oxygen-carrying capacity. Soot particles cause respiratory problems and contribute to air pollution.

Free Radical Substitution

Alkanes react with halogens (Cl_2, Br_2) in the presence of **UV light** via a free radical mechanism. The UV light provides the energy for **homolytic fission** of the Cl–Cl bond. This mechanism has three stages:

1. Initiation: UV light causes homolytic fission of the Cl–Cl bond, producing two chlorine radicals (each with an unpaired electron).

$$Cl_2 \rightarrow 2Cl\bullet \quad \text{(homolytic fission)}$$

2. Propagation: A chain reaction in which one radical reacts to produce another radical. Each propagation step generates a new radical that continues the chain.

$$Cl\bullet + CH_4 \rightarrow \bullet CH_3 + HCl$$

$$\bullet CH_3 + Cl_2 \rightarrow CH_3Cl + Cl\bullet$$

3. Termination: Two radicals combine, removing radicals from the system and ending the chain.

$$Cl\bullet + Cl\bullet \rightarrow Cl_2 \quad \bullet CH_3 + Cl\bullet \rightarrow CH_3Cl \quad \bullet CH_3 + \bullet CH_3 \rightarrow C_2H_6$$

Limitations of free radical substitution:

- Further substitution can occur (producing CH_2Cl_2, $CHCl_3$, CCl_4), so a **mixture of products** is obtained
- This makes it unsuitable for preparing a specific mono-substituted product in good yield
- The termination step $\bullet CH_3 + \bullet CH_3 \rightarrow C_2H_6$ explains the trace of ethane found in the products

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Exam Tip: Use the correct curly arrow notation: **single-headed (fishhook) arrows** for radical mechanisms showing the movement of one electron, and **double-headed arrows** for ionic mechanisms showing the movement of a pair of electrons. In initiation, always state "homolytic fission" and show the dot on the radical. You must show all three stages for full marks.

4.3 Alkenes

Alkenes: Unsaturated hydrocarbons containing at least one C=C double bond. The double bond consists of a sigma bond (head-on overlap) and a pi bond (sideways overlap of p orbitals). The pi bond creates an area of high electron density above and below the plane of the molecule, making alkenes susceptible to attack by **electrophiles** (electron-seeking species).

Test for Unsaturation

Add **bromine water** (orange/brown) to the sample and shake. If a C=C is present, the bromine water is **decolourised** (turns colourless) as Br_2 adds across the double bond. Alkanes do not decolourise bromine water (no reaction without UV light).

Electrophilic Addition Reactions

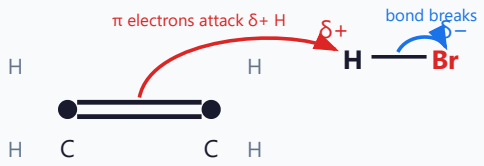
Electrophilic addition: A reaction in which an electrophile (an electron pair acceptor with a full or partial positive charge) is attracted to the electron-rich C=C double bond. The pi bond breaks and two new sigma bonds form. The product is a single saturated molecule.

Reagent	Conditions	Product	Notes
HBr	Room temperature, gaseous or concentrated solution	Bromoalkane	Markownikoff's rule applies for asymmetric alkenes
H_2O (steam)	H_3PO_4 catalyst, 300 °C, 60–70 atm	Alcohol	Industrial hydration of ethene to ethanol

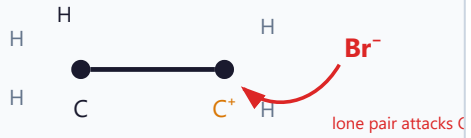
Br_2	Room temperature (bromine water or pure Br_2)	Dibromoalkane	Decolourises bromine water — test for C=C
H_2	Ni catalyst, 150 °C (hydrogenation)	Alkane	Used in margarine production to harden vegetable oils
HCl	Room temperature	Chloroalkane	Similar mechanism to HBr addition

Electrophilic Addition: Ethene + HBr

Step 1: Electrophilic attack



Step 2: Nucleophilic attack



Product: Bromoethane (CH_3CH_2Br)

Key Points for Mechanism Questions

1. Show dipole on HBr (δ+H–Brδ−)
2. Curly arrow from C=C π bond to δ+ H (electrophilic attack)
3. Curly arrow from H–Br bond to Br (heterolytic fission → Br^-)
4. Curly arrow from Br^- lone pair to C^+ carbocation (nucleophilic attack)

Use double-headed curly arrows showing movement of electron PAIRS

Markownikoff's Rule

When HBr adds to an asymmetric alkene (e.g. propene), the **major product** is the one where the H adds to the carbon already bonded to more hydrogen atoms (the less substituted carbon). This produces the **more stable** secondary (or tertiary) carbocation intermediate rather than the less stable primary carbocation. The greater stability arises from the **inductive (electron-donating) effect** of alkyl groups, which partially stabilises the positive charge by spreading it over a larger area.

Exam Tip: You need to explain WHY Markownikoff's rule gives the major product: it is because the more substituted carbocation intermediate is more stable, so it forms preferentially. Simply stating the rule without the stability explanation will not gain full marks at A2.

Addition Polymers

Addition polymerisation: Many small alkene molecules (monomers) join together to form a long-chain polymer by opening their C=C double bonds to form C–C single bonds. No by-product is formed — all atoms from the monomers end up in the polymer. Atom economy = 100%.

$$nCH_2{=}CH_2 \rightarrow (-CH_2-CH_2-)_n \quad \text{(poly(ethene))}$$

Addition polymers such as poly(ethene) and poly(propene) are **non-biodegradable** because the strong C–C backbone is resistant to chemical and biological attack. Disposal is a significant environmental concern. Options include: landfill (slow degradation), incineration (energy recovery but produces CO_2), mechanical recycling (sorting, melting, remoulding), and feedstock recycling (cracking back to monomers).

4.4 Alcohols

Classification

Classification	Structure	Example	Key Feature
Primary (1°)	OH group on a carbon attached to one other carbon (or none)	Butan-1-ol	Can be oxidised to aldehyde, then carboxylic acid
Secondary (2°)	OH group on a carbon attached to two other carbons	Butan-2-ol	Can be oxidised to ketone only
Tertiary (3°)	OH group on a carbon attached to three other carbons	2-methylpropan-2-ol	Resistant to oxidation

Oxidation of Alcohols

The oxidising agent is acidified potassium dichromate(VI), $K_2Cr_2O_7/H_2SO_4$. The colour change from **orange** $Cr_2O_7^{2-}$ to **green** Cr^{3+} confirms that oxidation has occurred. [O] is used above the arrow to represent the oxidising agent.

Alcohol Type	Product with Distillation	Product with Reflux
Primary	**Aldehyde** (distil immediately to prevent further oxidation — the aldehyde has a lower boiling point and is removed as it forms)	**Carboxylic acid** (full oxidation with excess oxidising agent)
Secondary	**Ketone** (no further oxidation possible; the carbonyl carbon has no H to lose)	
Tertiary	**No reaction** (resistant to oxidation under these conditions; the orange colour persists)	

Exam Tip: To produce an **aldehyde** from a primary alcohol, you must **distil the product immediately** as it forms (the apparatus is set up for distillation, not reflux). If you reflux instead, the aldehyde remains in contact with the oxidising agent and is further oxidised to a carboxylic acid. This distinction between distillation and reflux is a very common exam question.

Elimination (Dehydration)

Heating an alcohol with concentrated sulfuric acid (or concentrated phosphoric acid) removes water to form an alkene. This is an elimination reaction:

$$CH_3CH_2OH \longrightarrow [\text{conc. } H_2SO_4, \text{ heat}]\ CH_2{=}CH_2 + H_2O$$

Alternatively, passing alcohol vapour over hot aluminium oxide (Al_2O_3) at 300–400 °C achieves the same result. The acid or Al_2O_3 acts as a catalyst.

Esterification

An alcohol reacts with a carboxylic acid in the presence of a concentrated sulfuric acid catalyst (heated under reflux) to form an ester and water. This is a reversible, equilibrium reaction with limited yield:

$$CH_3COOH + CH_3CH_2OH \rightleftharpoons CH_3COOCH_2CH_3 + H_2O$$

Esters have characteristic sweet, fruity smells and are used as flavourings, fragrances, solvents, and plasticisers.

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4.5 Haloalkanes

Nucleophilic Substitution

Nucleophile: An electron pair donor — a species with a lone pair of electrons that can form a new covalent bond by attacking a positively charged or electron-deficient carbon atom. Examples: OH^-, NH_3, CN^-, H_2O.

The carbon bonded to the halogen is electron-deficient (δ+) because halogens are more electronegative than carbon. This δ+ carbon is the site of nucleophilic attack.

Nucleophilic Substitution Reactions of Haloalkanes

Hydrolysis: Haloalkane → Alcohol

Reagent: aqueous NaOH (warm) Nucleophile: OH^-

$CH_3CH_2Br + NaOH \rightarrow CH_3CH_2OH + NaBr$

Conditions: heat under reflux with aqueous NaOH

Nitrile formation: Haloalkane → Nitrile (+1 carbon)

Reagent: KCN in ethanol/water Nucleophile: CN^-

$CH_3CH_2Br + KCN \rightarrow CH_3CH_2CN + KBr$

Important: increases carbon chain length by 1

Amine formation: Haloalkane → Primary amine

Reagent: excess conc. NH_3 in ethanol, sealed tube, heat Nucleophile: NH_3

$CH_3CH_2Br + 2NH_3 \rightarrow CH_3CH_2NH_2 + NH_4Br$

Excess NH_3 ensures primary amine is the main product

S_N1 vs S_N2 Mechanisms

Feature	S_N2 (primary haloalkanes)	S_N1 (tertiary haloalkanes)
Number of steps	**One step** (concerted): nucleophile attacks as C–X bond breaks simultaneously	**Two steps**: (1) C–X bond breaks heterolytically to form a carbocation, (2) nucleophile attacks the carbocation
Rate-determining step	The single concerted step	Step 1: formation of the carbocation (slow)

Rate equation	Rate = k[haloalkane][nucleophile]	Rate = k[haloalkane] only
Why this substrate?	Primary carbon is less sterically hindered, allowing the nucleophile to approach	Tertiary carbocation is stabilised by 3 electron-donating alkyl groups (inductive effect)
Stereochemistry	Inversion of configuration (backside attack)	Racemic mixture (carbocation is planar; attack from either side)

Secondary haloalkanes can react via either mechanism depending on conditions (stronger nucleophile favours S_N2; polar protic solvent favours S_N1).

Reactivity of Haloalkanes

The rate of hydrolysis depends on the **strength of the C–X bond** (not electronegativity). Bond enthalpy decreases down the group: C–F (484 kJ mol^{-1}) > C–Cl (338) > C–Br (276) > C–I (238). Therefore, **iodoalkanes react fastest** and fluoroalkanes react slowest. The weaker the bond, the easier it is to break.

> **Exam Tip:** Students often confuse bond strength with bond polarity here. Although C–F is the most polar bond, fluoroalkanes react the *slowest* because the C–F bond is the strongest. It is the **bond enthalpy** (not polarity) that determines rate of hydrolysis.

Ozone Depletion

Chlorofluorocarbons (CFCs) were widely used as refrigerants, aerosol propellants, and foam-blowing agents. In the stratosphere, UV radiation breaks the C–Cl bond homolytically to produce chlorine radicals that **catalytically destroy ozone**:

$$Cl\bullet + O_3 \rightarrow ClO\bullet + O_2$$

$$ClO\bullet + O_3 \rightarrow Cl\bullet + 2O_2$$

The chlorine radical is regenerated, so **one CFC molecule can destroy thousands of ozone molecules**. This is why Cl• acts as a catalyst. The Montreal Protocol (1987) banned CFCs, and they have been replaced by HCFCs (which break down in the lower atmosphere) and HFCs (which contain no chlorine).

4.6 Organic Synthesis

Organic synthesis involves planning a series of reactions to convert a starting material into a desired product. You should be able to suggest suitable reagents and conditions for each step, recognise how different functional groups can be interconverted, and plan multi-step syntheses. The key reaction pathways connect: alkanes ↔ haloalkanes ↔ alcohols ↔ aldehydes/ketones ↔ carboxylic acids, with alkenes as a branching point.

Exam Tip: Learn the key reaction pathways as a "roadmap" connecting all functional groups. For each transformation, know: (1) the reagent(s), (2) the conditions (temperature, solvent, catalyst), (3) the type of reaction (addition, substitution, elimination, oxidation, reduction). Be prepared to describe multi-step syntheses with full reagents and conditions at each step. If asked to increase the carbon chain length, use the CN^- nucleophile (KCN in ethanol, reflux).

4.7 Analytical Techniques (AS)

Mass Spectrometry

A mass spectrometer determines the relative molecular mass (M_r) of a compound. The **molecular ion peak (M^+)** has the highest m/z value (ignoring the small M+1 isotope peak) and corresponds to the intact molecule that has lost one electron. Fragmentation patterns provide information about the structure.

m/z Difference from M^+	Fragment Lost	Likely Group
15	CH_3•	Methyl group
17	OH•	Hydroxyl group
18	H_2O	Alcohol (loss of water)
29	CHO• or C_2H_5•	Aldehyde or ethyl group
45	OC_2H_5•	Ethoxy group (ester)

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Infrared (IR) Spectroscopy

IR spectroscopy identifies functional groups by measuring which frequencies of infrared radiation are absorbed by a molecule. Different covalent bonds absorb at characteristic wavenumbers because they vibrate (stretch and bend) at specific frequencies.

Bond	Wavenumber / cm^{-1}	Absorption Appearance
O–H (alcohol)	3230–3550	Broad
O–H (carboxylic acid)	2500–3300	Very broad (often overlaps C–H)
N–H	3300–3500	Medium, sharp (2 peaks for primary amine)
C–H	2850–3100	Medium
C=O	1680–1750	Strong, sharp
C=C	1620–1680	Medium
C–O	1000–1300	Strong

The **fingerprint region** (below 1500 cm^{-1}) is unique to each compound and can be matched against a database for positive identification, like a molecular "fingerprint." You will be given a data table in the exam.

Exam Tip: In IR questions, look for the **presence or absence** of key peaks: a broad O–H around 3300 cm^{-1} indicates an alcohol; a very broad O–H stretching over 2500–3300 cm^{-1} indicates a carboxylic acid; a sharp C=O around 1700 cm^{-1} indicates a carbonyl compound. The disappearance or appearance of peaks can **confirm a reaction has occurred** (e.g. loss of broad O–H and appearance of C=O confirms oxidation of an alcohol to an aldehyde/ketone).

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